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Scale considerations for integrating forest inventory plot data and satellite image data for regional forest mapping

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ABSTRACT

The integration of satellite image data with forest inventory plot data is a popular approach for mapping forest vegetation over large regions. Several methodological choices regarding spatial scale, mostly related to spatial resolution or grain, can profoundly influence forest maps developed from plot and imagery data. Yet often the consequences of scaling choices are not explicitly addressed. Our objective was to quantify the effects of several scale-related methods on map accuracy for multiple forest attributes, using a variety of diagnostics that address different map characteristics, to help guide map developers and users. We conducted nearest-neighbor imputation over a large region in the Pacific Northwest, USA, to investigate effects of imputation grain (single pixel or kernel); inclusion of heterogeneous plots; accuracy assessment grain and extent; and value of k ($k = 1$ and $k = 5$), where k is the number of nearest-neighbor plots. Spatial predictors were from rasters describing climate and topography and a time-series of Landsat imagery. Reference data were from regional forest inventory plots measured over two decades. All analyses were conducted at a spatial resolution of 30 m × 30 m. Effects of imputation grain and heterogeneous plots on map accuracy were small. Excluding heterogeneous plots slightly improved map accuracy and did not lessen the systematic agreement (AC_{SYS}) between our maps and observed plot data. Accuracy assessment grain strongly influenced map accuracy: maps assessed with a multi-pixel block were much more accurate than when assessed with a single pixel for almost all map diagnostics, but this was an artifact of methods rather than reflecting real differences among maps. Unsystematic agreement (AC_{UNS}) between our maps and plots, or random error, improved notably with increasing accuracy assessment extent for all scaling methods, indicating that reliability of most map applications can be improved through coarsening the map grain. Value of k strongly influenced map diagnostics. The $k = 5$ maps were better than $k = 1$ maps for local-scale accuracy, but at the cost of reduced AC_{SYS}, and loss of variability and poor areal representation of forest conditions over the study region. The $k = 1$ maps produced notably better predictions of the least abundant forest conditions (early-successional, late-successional, and broadleaf). None of the scaling methods were optimal for all map diagnostics. Nevertheless, given a variety of diagnostics associated with a range of scaling options, map developers and map users can make informed choices about methods and resulting maps that best meet their particular objectives, and we present some general guidelines in this regard. Most of our findings are applicable to mapping with Landsat data in other forested regions with similar forest inventory data, and to other methods for spatial prediction.

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1. Introduction

Applications of remote sensing to problems in ecology and in land and environmental management have grown exponentially in recent years. In particular, remotely sensed data acquired by the Landsat sensors have played a key role in ecological applications (Cohen & Goward, 2004), vegetation mapping (Xie, Sha, & Yu, 2008), and broad-scale forest inventory (McRoberts, Tomppo, & Næsset, 2010; Tomppo et al., 2008). Landsat imagery offers moderate spatial resolution, a long history, and a data archive that is accessible at no cost (Woodcock et al., 2008). These factors provide unique opportunities to extend applications of Landsat time-series to monitoring of forest change (Kennedy, Yang, & Cohen, 2010; Kennedy et al., 2012; Ohmann et al., 2012).

Despite its acknowledged utility, remotely sensed data cannot completely replace ground sample data for many applications. The information needs for broad-scale land cover data have expanded to include biodiversity, dead wood, species composition, and other attributes that cannot yet be reliably sensed remotely. Consequently, the integration of satellite imagery with regional or national forest inventory plot data has become a popular approach for estimation and spatial prediction of forest attributes over large geographic regions (McRoberts et al., 2010; Tomppo et al., 2008). Many countries in the world now use sample-based approaches for national forest inventories, and remote sensing is

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often used to enhance sampling and estimation through stratified, model-assisted, or model-based estimation (McRoberts et al., 2010). Many applications in ecology and ecosystem science also have combined remotely sensed and plot data for model-based spatial prediction over large landscapes.

With any spatial modeling, mapping, or estimation approach, the use of field plot data as training data or for model validation requires geographical co-registration of plot locations within imagery, so that paired values can be obtained from the two datasets. This co-registration raises a host of methodological issues having to do with scale, particularly spatial resolution (grain). Yet often the consequences of scaling choices, which interact with the spatial heterogeneity of the vegetation being mapped, are not addressed explicitly by the investigators (but see Arponen, Lehtomaki, Leppanen, Tomppo, & Moilanen, 2012; Fassnacht, Cohen, & Spies, 2006; McRoberts, 2010; Stehman & Wickham, 2011). A review by Lechner, Langford, Bekessy, and Jones (2012) reports that sources of uncertainty in mapping analyses – including several scale-dependent factors – were rarely addressed in landscape ecology studies that used spatial data.

Our study objective was to quantify the effects of several scale-related methods on estimates of map accuracy for multiple forest attributes. We define accuracy based on a variety of diagnostics that address different map characteristics. Although our method for vegetation mapping in this study was nearest-neighbor imputation (Eskelson et al., 2009), most of the scale issues we addressed apply more broadly to other methods for predictive vegetation mapping that rely on integrating plot and geospatial data. Our overall approach was to develop multiple versions of forest vegetation maps, using different combinations of scaling options, and then compare the results for a variety of map diagnostics. Implicit to our analysis is the concept that regional, multi-attribute forest maps serve a wide range of user needs, that there are trade-offs among different mapping approaches, and that no single spatial unit (e.g. pixels, blocks of pixels, polygons) is universally best for mapping or for accuracy assessment (Stehman & Wickham, 2011). We also recognize that relative model performance will vary to some degree among forest attributes, for different diagnostics, and among different ecosystems. Nevertheless, we draw some general conclusions from our findings that can inform both developers and users of regional forest maps developed from regional inventory plots and satellite imagery.

Nearest-neighbor techniques have emerged within the international forestry community as useful methods for predicting forest attributes as combinations of the k observations (i.e. field plots) that have similar characteristics in a space of ancillary variables (McRoberts, 2012; McRoberts et al., 2010; Tomppo et al., 2008). Nearest-neighbor methods are appealing because they are multivariate and nonparametric (require no assumptions about the distributions of response or predictor variables), and can be used to map multiple forest characteristics over large areas (Eskelson et al., 2009; McRoberts, 2012). Nearest-neighbor techniques based on forest inventory plots and satellite imagery were first implemented operationally in Finland in 1990, but have now been applied in locations spanning the globe (McRoberts et al., 2010).

Gradient nearest neighbor (GNN) is one variation of nearest-neighbor imputation that relies on constrained ordination (direct gradient analysis) for weighting distances in nearest-neighbor calculations (Ohmann & Gregory, 2002) (see Section 2.1). In past studies we have used GNN to map forest vegetation for a single point-in-time, for a variety of forest ecosystems and objectives (Ohmann, Gregory, Henderson, & Roberts, 2011; Ohmann, Gregory, & Spies, 2007; Pierce, Ohmann, Wimberly, Gregory, & Fried, 2009). We recently extended GNN to mapping multiple forest attributes at two dates based on two dates of Landsat imagery (Ohmann et al., 2012). For the current effort, conducted as part of a larger study to integrate data from Landsat time-series and regional inventory plots in an observation-based system for biomass and carbon monitoring in wooded ecosystems, we more fully utilized the Landsat time-series data to map a yearly time-series. Although we use multi-temporal plot and imagery data in our analyses, this is not a primary emphasis of this paper, and our methods and findings are equally applicable to single-date data. In addition, although we assessed the scaling effects empirically for the Oregon and California Cascades in the northwestern USA (Fig. 1), this region encompasses a wide range of physical environments and forest vegetation and our results should be generalizable to other locations.

We considered several aspects of spatial scale, which are described in more detail below: the imputation grain (the scale of the mapping unit used for nearest-neighbor distance calculations); the vegetation heterogeneity within plots used in modeling (which interacts with spatial grain); the accuracy assessment grain (single pixel or multi-pixel block); the spatial extent of accuracy assessment (from plot to larger hexagons); and the value of k, which refers to the number of nearest-neighbor plots used in calculating the forest attribute value that is imputed to a map unit. The effect of k is unique to nearest-neighbor imputation. Because we used Landsat satellite imagery, all analyses were conducted using raster data at a spatial resolution of 30 m $\times$ 30 m (referred to as 30-m).

2. Methods

2.1. GNN process for gradient modeling and spatial prediction

We developed a time-series of GNN maps for each combination of scaling options (imputation grain, within-plot heterogeneity, accuracy assessment grain and extent, and value of k). GNN was implemented as described in Ohmann and Gregory (2002) and Ohmann et al. (2011, 2012), but with the addition of kernel imputation, $k = 5$, and enhancements for multi-date (yearly) mapping. Neighbor selection in GNN is based on weighted Euclidean distance within multivariate gradient space as determined from canonical correspondence analysis (CCA) (ter Braak, 1986), a method of constrained ordination (direct gradient analysis).

Spatial predictors (explanatory variables, often referred to collectively as feature space) are listed in Table 1. Spectral variables were derived from Landsat imagery mosaics developed with the LandTrendr (Landsat Detection of Trends in Disturbance and Recovery) algorithms (Kennedy, Cohen, & Schroeder, 2007; Kennedy et al., 2010). LandTrendr is a trajectory-based change detection method that simultaneously examines a time-series of yearly Landsat TM satellite images. Using images

Fig. 1. The Oregon and California Cascades modeling region, shown in darker gray.

Table 1
Spatial predictors used in CCA and GNN imputation. All rasters were 30 m resolution.

Variable subset	Code	Description
Landsat time-series, processed using LandTrendr algorithms (Kennedy et al., 2010)	*TC1*	Axis 1 (brightness) from tasseled cap transformation (Crist & Cicone, 1984).
	TC2	Axis 2 (greenness) from tasseled cap transformation.
	TC3	Axis 3 (wetness) from tasseled cap transformation.
Climate, from PRISM rasters (Daly et al., 2008), which are 30-year normals	*ANNTMP*	Mean annual temperature (°C).
	AUGMAXT	Mean maximum temperature in August (°C).
	DECMINT	Mean minimum temperature in December (°C).
	ANNPRE	Mean annual precipitation (natural logarithm, mm).
	CONTPRE	Percentage of annual precipitation falling from June–August, a measure of continentality.
	SMRTP	Growing season moisture stress, the ratio of mean temperature (°C) to precipitation (natural logarithm, mm) from May–September.
	COASTPROX	Index of coastal proximity, based on penetration of marine air through complex terrain as indicated by minimum and maximum temperatures (index from 0 to 1000).
Topography, developed from 10-m digital elevation model (DEM) and rescaled to 30 m.	*ELEV*	Elevation (m).
	ASP	Cosine transformation of aspect (degrees).
	SLP	Slope (%).
	SOLAR	Cumulative potential relative radiation during the growing season (Pierce, Lookingbill, & Urban, 2005).
	TPI	Topographic position index, calculated as the difference between a cell's elevation and the mean elevation of cells within a 450-m-radius window.
Location	*LAT*	Geographic latitude (degrees).
	LONG	Geographic longitude (degrees).

that are cloud-free, geometrically corrected, and radiometrically normalized (Kennedy et al., 2010), the LandTrendr algorithms identify fitted line segments that are of consistent trajectory for each pixel that describe sequences of disturbance and growth. The algorithms minimize annual variability from differences in sun angle, phenology, and atmospheric effects, such that the remaining signal more closely reflects real changes in vegetation. The "temporally smoothed" pixel data are mosaicked across scenes to provide consistent coverage over larger regions. We used data from these LandTrendr-derived "temporally smoothed" imagery mosaics, which consisted of an annual time-series of mosaics from 1984 to 2011, as spatial predictors. We used the tasseled cap indices (Crist & Cicone, 1984) as explanatory variables, rather than the spectral variables, because they capture most of the important variation in forest structure (Cohen & Goward, 2004) using fewer variables.

Response variables for CCA were basal area by species and size-class observed on 8670 field plots installed at 4391 locations, measured from 1991 to 2010 in regional and national forest inventories: Forest Inventory and Analysis (FIA) Annual Inventory (Bechtold & Patterson, 2005), FIA periodic inventories by the U.S. Forest Service (USFS) Pacific Northwest Research Station (PNW) and Region 5 (Waddell & Hiserote, 2005), and Current Vegetation Survey (CVS) by USFS Region 6 and BLM (Max et al., 1996). Detailed descriptions of sample designs and field methods can be found in these references. FIA Annual Inventory plots consist of a cluster of four subplots, each containing nested plots for sampling trees of different diameter classes (Fig. 2). We used data from all nested plots, including the macroplot, a measurement option employed in the Pacific Northwest because of the prevalence of very large trees. At each plot location there were as many as three separate field measurements,

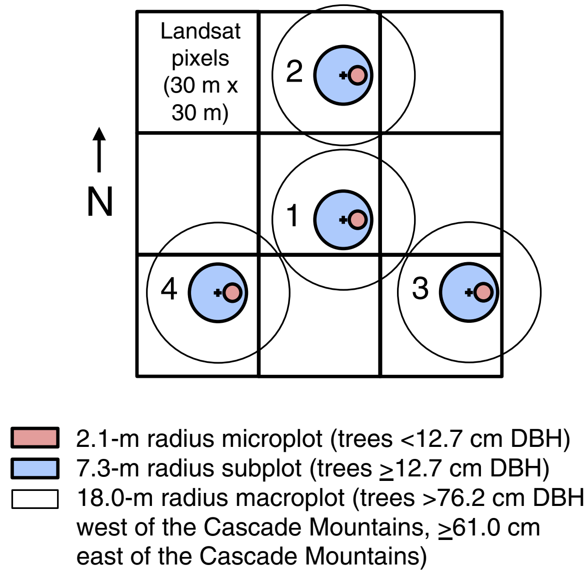

Fig. 2. Layout of FIA Annual Inventory plot comprised of four subplots, and showing 3-by-3-pixel block that provides associated values for spatial predictors.

conducted at different times. Some were remeasurements using the same configuration and measurement protocols (e.g. CVS plots were remeasured at irregular time intervals, from 1 to 14 years), but others differed (e.g. FIA Annual Inventory plots could be installed at a new location, at the same location as a previous FIA periodic plot, or at the same location as a CVS plot).

We used only those plots that were at least 50% forest land (land that is at least 10% stocked by forest trees of any size, or land formerly having such tree cover, and not currently developed for a nonforest use, and meeting minimum area criteria; Bechtold & Patterson, 2005). For each plot, tree-level data from all subplots were combined, and converted to per-hectare values that described the forest portion of the plot, which was treated as homogenous. We did not average conditions across forest and nonforest land classes within a plot, because our modeling and mapping approach applied only to the forested component of the landscape. Imputation was applied to pixels of all land classes, but nonforest areas were masked in the final vegetation map.

To determine values for spatial predictors associated with each plot observation, we used a 3-by-3-pixel block that encompasses the outer extent of the general area sampled by the field plot (Fig. 2). We intersected the 3-by-3-pixel block with the LandTrendr mosaic for the same year as plot measurement, and the means of the tasseled cap indices for the nine pixels were associated with the plot observation. Values for the other explanatory variables were similarly extracted for the plot footprints, but values were the same for all years (e.g., climate was assumed constant over the range of plot measurement dates).

We developed two CCA models, based on two sets of reference plots: one that included heterogeneous plots that encompassed strongly contrasting land cover classes or forest conditions ($n = 8670$), and one that excluded them ($n = 6036$). Heterogeneous plots were identified by viewing outliers from the GNN maps in the LandTrendr imagery and in digital aerial photography. We also screened plots for disturbances occurring between the plot assessment date and the imagery date, such as harvest or fire, where the spectral signal would not be representative of the plot data. We found very few of these situations because of the yearly temporal matching. Plots of all measurement years were used in the CCA models. The validity of this approach relies on the assumption that LandTrendr effectively normalizes the spectral values among image dates, i.e. that spectral values representing similar forest conditions were equivalent across the 28 imagery years.

The two CCA models (with and without heterogeneous plots) were then used in imputation at two grains (single-pixel and kernel) and for two values of k ($k = 1$ and $k = 5$), for each year in the 1984–2011 time-series. Neighbor-finding was based on weighted Euclidean distance, where the weights were obtained using CCA (CCA axes scores weighted by their eigenvalues). Distances were calculated using the quadratic form, which is most widely used for nearest-neighbor imputation (Stage & Crookston, 2007):

$$d^2_{ij} = \left(\mathbf{X}_i - \mathbf{X}_j\right)\mathbf{W}\left(\mathbf{X}_i - \mathbf{X}_j\right)'$$

where i denotes a target set element (map pixel) for which a prediction is sought, j denotes a reference set (plot) element, $\mathbf{X}_i$ and $\mathbf{X}_j$ are $(1 \times p)$ vectors of observations of explanatory (or feature space) variables for the ith and jth elements, respectively, and $\mathbf{W}$ is a square matrix. Our $\mathbf{W}$ (weights) is a square matrix of size $p \times p$ from CCA:

$\mathbf{CEC}'$

where: $\mathbf{C}$ coefficient matrix from CCA ($p \times a$), $\mathbf{E}$ diagonal eigenvalue matrix from CCA ($a \times a$), $\mathbf{C}'$ transpose of C ($a \times p$), p number of explanatory variables, and a is the number of CCA axes.

2.2. Scaling options considered in this study

We did not explicitly address the grain of the field plot data in our analyses. The regional forest inventory plots consist of a cluster of four (FIA Annual Inventory) or five (all periodic inventories) subplots distributed over about 1 ha (Fig. 2). While others have used individual subplots as the observations (e.g., McRoberts, 2009), we chose to use whole plots for several reasons. First, plot locations in our region have been shown (unpublished data) to be insufficiently reliable for matching a single subplot to a single 30-m pixel, which is most important for Landsat and topographic predictors that vary at a fine spatial resolution, and for the fine-scale variability that characterizes much of the forest in this region. Although plot location errors influence results at all spatial grains, smaller spatial units (subplots and individual pixels) are more sensitive than larger units (whole plots and blocks of pixels) (Stehman & Wickham, 2011). Second, spectral reflectance values associated with a single 30-m Landsat pixel indicate conditions over an area that is substantially larger than a subplot. In addition, the subplot typically is not centered on the pixel. Third, for many of the older FIA periodic plots used in our modeling, the number of trees tallied at the subplot level is insufficient to characterize forest conditions and the subplot data were never intended to be analyzed separately from the entire plot (MacLean, 1980).

2.2.1. Imputation grain

We performed imputation at the single-pixel scale and for a 3-by-3-pixel kernel, or moving window, which we refer to as "kernel imputation." For single-pixel imputation, distances were calculated from explanatory variables and CCA scores for each individual pixel. In kernel imputation, CCA scores were calculated from the mean values of each explanatory variable for the nine pixels in a 3-by-3 window centered on each pixel. Nearest-neighbor distances were then calculated for the center (focal) pixel in the same way as for single-pixel imputation (Fig. 3).

2.2.2. Within-plot heterogeneity

We also evaluated the effect on our spatial predictions of excluding plots that encompass multiple forest conditions (e.g. older forest and

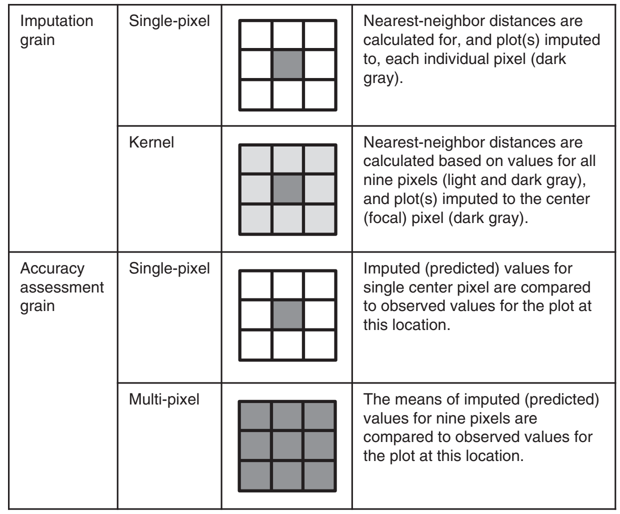

Fig. 3. Configurations of 30-m pixels for two imputation grains and two accuracy assessment grains. Shading indicates pixels whose values are used in each operation.

recently harvested forest, mixtures of forest and nonforest), which we refer to in this paper as "within-plot heterogeneity." The choice of whether to exclude heterogeneous, multi-condition plots (Ek & Burk, 1995; Fassnacht et al., 2006) is inherently scale-related because it requires addressing the degree of within-plot variability that is acceptable for modeling and mapping. In all of our analyses, we treated the plot data as homogenous. In other words, we did not explore options for explicitly accommodating heterogeneous plots, such as by assigning subsets of map pixels to different plot components.

Within-plot heterogeneity applies to both the plot data and the spatial data (spatial predictors), and is a function of the area encompassed by the plot and its interaction with variability in the spatial predictors within the 3-by-3-pixel block centered on the plot (Figs. 2 and 3). Although theoretically we could apply quantitative rules to both plot and spatial data to identify heterogeneous plots as those that exceed a heterogeneity threshold, in practice it has proven to be extremely difficult to achieve desired results in any way other than viewing each plot within the imagery and applying qualitative guidelines as objectively as possible. Guidelines were designed to approximate the FIA Annual Inventory rules for identifying separate condition classes (Bechtold & Patterson, 2005).

Researchers in remote sensing and ecology sometimes avoid this issue by selecting homogenous training observations or field plots, and making spatial predictions at the scale of the individual pixel (which by definition is homogenous). But this is not an option when using regional forest inventory plots that are installed in a fixed configuration regardless of ground conditions. In studies using FIA plots where condition classes are identified in the field, some researchers have confined their analyses to single-condition plots (e.g., Blackard et al., 2008). This approach runs the risk of influencing an accuracy assessment towards homogenous areas that are likely to be mapped with greater confidence, and missing fragmented portions of the landscape (Riemann, Wilson, Lister, & Parks, 2010). We addressed this potential problem in this study by computing many of the diagnostics with plot datasets that included heterogeneous plots.

2.2.3. Accuracy assessment grain

Observed forest attribute values for plots were compared to predicted values for a single pixel centered on the plot, and for a multi-pixel block (3-by-3-pixel block), where predicted values were computed as the mean of the predicted values for the nine pixels centered on the plot (Fig. 3). Although kernel imputation combined with multi-pixel accuracy assessment technically reflects conditions over a larger block (5-by-5 pixels), the nearest-neighbor distance measurements are heavily weighted towards the nine central pixels, and the calculation of predicted attribute values is constrained to the nine pixels that represent the plot footprint, which most closely corresponds to field measurements.

2.2.4. Spatial extents for summarizing accuracy assessment data

Vegetation maps are most commonly assessed for accuracy locally, at the spatial scale (resolution) of the individual plot observation (in our case, a 3-by-3 block of pixels centered on the plot; see Figs. 2 and 3). Yet for many map applications, mapped data are summarized and used at larger geographic extents, and map reliability assessed for these larger areas may be more relevant than plot-level accuracy (Riemann et al., 2010). In this study, we evaluated map accuracy at the plot level, and for all plots whose centers fell within larger cells of hexagonal tessellations. We generated two polygon datasets, providing two scales (extents) of equal-area tesselated hexagons. In the first dataset, hexagons were generated around centroids spaced 10 km apart, resulting in n = 453 hexagons of 8660 ha in size. Hexagons in the second dataset were spaced 30 km apart, resulting in n = 72 hexagons of 78,100 ha in size. For convenience, we refer to these hexagon tessellations as "10-km hexagons" and "30-km hexagons." The 10-km hexagons correspond to the approximate size of the smallest counties in the USA, and contain the minimum number of plots used by FIA to define its strata for estimation.

2.2.5. Value of k

Nearest-neighbor imputation uses either a single neighbor plot ($k = 1$), or a weighted average (or other function) of field observations from $k > 1$ neighbors (Eskelson et al., 2009). In this study we compared results for just two values of k, $k = 1$ and $k = 5$. For $k = 5$ we used inverse distance weighting to the target pixel to provide weights for neighbor observations for forest attributes. We analyzed just two values of k, rather than a larger range of values, because our primary purpose was to illustrate typical differences in maps based on $k = 1$ and $k > 1$, rather than to present an exhaustive analysis of the effects of k. Although our previous published research using GNN has used $k = 1$ (e.g., Ohmann et al., 2012), much of the published kNN research uses $k > 1$. In practice, the optimal choice of k is difficult to determine and depends on the criterion used in the evaluation, and may range from <3 to 10 or much greater (Eskelson et al., 2009).

2.3. Accuracy assessment

For accuracy assessment of the GNN maps, we computed a suite of diagnostics. Some were based on modified leave-one-out (LOO) cross-validation (Ohmann & Gregory, 2002), and others quantify agreement between the target pixel predictions and observations from the reference plots (the systematic sample of FIA Annual Inventory plots) (Riemann et al., 2010). We also computed the diagnostics from a completely independent validation dataset, but because results were so similar to LOO cross-validation we do not present them here.

For accuracy assessment, we did not consider observations for the plots measured at the same location to be spatially nor temporally independent. When pairing observed and predicted values for LOO cross-validation, predicted values were from the k nearest independent plot(s), i.e. plots that were not at the same location as the observed plot. This procedure also was followed by Wilson, Lister, and Riemann (2012). The LOO cross-validation set contained heterogeneous plots only when they were included in the model being assessed.

Using paired predicted and observed values for the LOO cross-validation datasets, we computed diagnostics for nine continuous forest attributes (Table 2) that were chosen to represent a variety of measures of forest structure and composition, and for a vegetation classification (Table 3). The vegetation classes were determined based on rules applied to values for three continuous forest attributes (canopy cover (*CANCOV*), broadleaf proportion (*BA-BLF*), and average tree size (*QMD*); see Table 2). Predicted values for forest attributes were taken from the spatial predictions for the same map year as field plot measurement.

Table 2
Continuous forest attributes used in computing map diagnostics.

Forest attribute	Description
BA-BLF	Proportion of total basal area (*BA*) comprised of broadleaf trees.
BA	Total basal area (m^2/ha) of all trees ≥2.5 cm diameter at breast height (DBH).
BIOMASS	Total biomass (kg/ha) of all trees ≥2.5 cm DBH.
QMD	Quadratic mean DBH (cm) of trees classified as dominant or co-dominant in the field, a measure of the tree crown's access to light.
MEANDBH	Mean DBH (cm) of all live trees weighted by tree basal area, which emphasizes larger (overstory) trees.
STPH	Density (trees/ha) of standing dead trees (snags) ≥25 cm DBH and ≥5 m tall.
DVPH	Volume (m^3/ha) of all down wood ≥25 cm large end diameter and ≥3 m long.
CANCOV	Canopy cover (percent) of all live trees, calculated from tree tally based on species, DBH, height, live crown ratio, and stand density using allometric equations.
OGSI	Old-growth structure index (Spies et al., 2007), a composite index (0–100) based on stand age, density of trees ≥100 cm diameter at breast height (DBH), diversity of tree sizes, density of large snags, and down wood volume (see Ohmann et al. (2012) for more detail.)

We also assessed map accuracy using the protocol outlined by Riemann et al. (2010), which allowed us to evaluate model reliability across a range of spatial extents, and in a way that recognizes that error exists in both predicted values and observed values. These diagnostics were computed for continuous variables only, not for vegetation class. We computed diagnostics for plots, 10-km hexagons and 30-km hexagons. For each extent, we compared predicted (imputed) values to observed values for a systematic sample of locations centered on n = 2180 FIA Annual Inventory plots in the study region, each measured once from 2001 to 2010. The FIA Annual Inventory validation dataset used for this assessment contained heterogeneous plots. The 10-km hexagons contained an average of 3.2 FIA Annual Inventory plots, and the 30-km hexagons contained an average of 21.1 plots. Observed values were for individual FIA Annual Inventory plots (at the plot scale), or computed as the mean of all observed values for FIA Annual Inventory plots within each 10-km or 30-km hexagon. Predicted values were based on observations for the k-nearest independent reference plots (plot scale), or on the means of observed values imputed to all FIA plot locations within the hexagons (10-km and 30-km hexagon extents).

For each of the three extents, we calculated the symmetric geometric mean functional relationship (GMFR) regression line (Ricker, 1984) from a scatterplot of the predicted and observed values. Unlike least squares regression, the GMFR is a symmetric regression model that describes the relationship between two datasets that are both subject to error. From this relationship, we calculated the agreement coefficients for systematic differences (AC_{SYS}) (differences in slope from the 1:1 line) and for unsystematic differences (AC_{UNS}), which reflect random error or lack of precision (scatter around the 1:1 line) (Ji & Gallo, 2006; Riemann et al., 2010). A value of 1.0 for AC_{SYS} or AC_{UNS} represents perfect agreement between two datasets.

3. Results

3.1. Overall map accuracy across scaling options

Local- (plot-) scale prediction accuracy varied widely among the nine forest attributes (Table 4), which represented a variety of measures of forest structure (Table 2). Across all scaling options, map accuracy generally was best for canopy cover (*CANCOV*) (average RMSE 0.24); intermediate for measures of total basal area, biomass, and average tree size (*BA*, *BIOMASS*, *QMD*, *MEANDBH*) (average RMSEs 0.42–0.64); and worst for measures of dead wood (*STPH* and *DVPH*) and broadleaf

Table 3
Vegetation classification used in computing map diagnostics. See Table 2 for continuous variable definitions.

Vegetation class	Definition	Abbreviation	'Fuzzy' correct classes
Sparse	*CANCOV* < 10	Sparse	Open
Open	*CANCOV* ≥ 10 and *CANCOV* < 40	Open	Sparse
Broadleaf, small	*CANCOV* ≥ 40, *BA-BLF* ≥ 0.65, *QMD* < 25	Blf-Sm	Open, Mix-Sm, Con-Sm
Broadleaf, medium/large	*CANCOV* ≥ 40, *BA-BLF* ≥ 0.65, QMD ≥ 25	Blf-Md/Lg	Blf-Sm, Mix-Md, Mix-Lg/VLg
Mixed conifer-broadleaf, small	*CANCOV* ≥ 40, *BA-BLF* ≥ 0.20 and *BA-BLF* < 0.65, *QMD* < 25	Mix-Sm	Open, Blf-Sm, Mix-Md, Con-Sm
Mixed conifer-broadleaf, medium	*CANCOV* ≥ 40, *BA-BLF* ≥ 0.20 and *BA-BLF* < 0.65, *QMD* ≥ 25 and *QMD* < 50	Mix-Md	Blf-Md/Lg, Mix-Sm, Mix-Lg, Con-Md
Mixed conifer-broadleaf, large/very large	*CANCOV* ≥ 40, *BA-BLF* ≥ 0.20 and *BA-BLF* < 0.65, *QMD* ≥ 50	Mix-Lg/VLg	Blf-Md/Lg, Mix-Md, Con-Lg, Con-VLg
Conifer, small	*CANCOV* ≥ 40, *BA-BLF* < 0.20, *QMD* < 25	Con-Sm	Open, Mix-Sm, Con-Md
Conifer, medium	*CANCOV* ≥ 40, *BA-BLF* < 0.20, *QMD* ≥ 25 and *QMD* < 50	Con-Md	Mix-Md, Con-Sm, Con-Lg
Conifer, large	*CANCOV* ≥ 40, *BA-BLF* < 0.20, *QMD* ≥ 50 and *QMD* < 75	Con-Lg	Mix-Lg/VLg, Con-Md, Con-VLg
Conifer, very large	*CANCOV* ≥ 40, *BA-BLF* < 0.20, *QMD* ≥ 75	Con-VLg	Mix-Lg/VLg, Con-Lg

tree abundance (*BA-BLF*) (average RMSEs 1.32–2.35). This pattern was expected given our reliance on Landsat imagery, which provides data on reflected sunlight from the canopy surface. Attributes of the forest canopy (e.g. *CANCOV*) are more directly correlated with canopy light reflectance, whereas attributes of the understory (e.g. *DVPH*) are hidden by the canopy and map accuracy relies on correlation with overstory elements. Whereas broadleaf abundance (*BA-BLF*) is relatively easy to predict in closed-canopy forest, the calculations of RMSE include observations in open-canopy forest where the spectral data are strongly influenced by bare soil and non-tree vegetation. Interestingly, map accuracy for the old growth structure index (*OGSI*), which is a composite index of several measures of live and dead trees, was similar to the accuracies for individual variables of intermediate accuracy (average RMSE 0.48).

Overall map accuracy for an 11-class vegetation classification ranged from 39 to 52% across the scaling options, but the maps were 81 to 90% accurate within one cover class, species composition class, or size class (Fig. 4, Appendix A) (see Table 3 for vegetation classes considered to be 'fuzzy' correct). Users and producers accuracies and kappa statistics (Cohen, 1960) for individual vegetation classes for the 16 scaling options are shown in the Appendix A.

In general, maps developed using $k = 1$ more closely matched the area distributions of vegetation classes present in the observed (plot) data than maps from $k = 5$, regardless of imputation grain (Fig. 5).

For the scaling options we evaluated, accuracies were most strongly associated with accuracy assessment grain and value of k (Figs. 4 and 6). Systematic agreement (AC_{SYS}) was quite good (0.78–0.97) and relatively consistent across all of the scaling options (Fig. 6). Unsystematic agreement (AC_{UNS}) ranged from poor (<0.00) when assessed at the plot extent, to somewhat better (0.16–0.50) at the 10-km hexagon extent, to relatively good (0.73–0.85) at the 30-km hexagon extent (Fig. 6). The geometric mean functional relationship (GMFR) regression lines are shown for one of the forest attributes (OGSI) as an example (Fig. 7a). We illustrate with OGSI because it is a composite index of several elements of forest structure, and older forest habitat is of interest to many map users.

3.2. Imputation grain

Differences in map accuracy associated with *imputation grain* were relatively small, and varied with accuracy assessment grain. Averaged for the nine forest attributes, normalized root mean square errors (RMSEs) (Fig. 4) and unsystematic agreement (AC_{UNS}) at the plot scale (Fig. 6) were slightly better for single-pixel than for kernel imputation when assessed using a multi-pixel block. However, kernel imputation yielded better results than single-pixel when assessed at the single-pixel grain. The same patterns were seen for accuracy of 11 vegetation classes (Fig. 4).

Systematic agreement (AC_{SYS}) at the plot scale was very similar for single-pixel and kernel imputation, with all AC_{SYS} values ≥0.78 (Fig. 6).

3.3. Effect of heterogeneous plots

For both imputation grains, both accuracy assessment grains, and both values of k, RMSEs and vegetation class accuracies were slightly better for maps where heterogeneous plots were excluded (Fig. 4). Differences were most pronounced for RMSEs and for $k = 1$. Maps with no heterogeneous plots also had better unsystematic agreement (AC_{UNS}) at the plot scale (Fig. 6). Excluding heterogeneous plots had a

Table 4
Root mean square errors (RMSEs) normalized by dividing the RMSE by the observed mean value for continuous forest attributes (see Table 2 for definitions), for GNN maps developed with different scaling options: value of k, imputation grain (kernel or single-pixel), with (Y = Yes, n = 8670) and without (N = No, n = 6036) heterogeneous plots, and accuracy assessment grain (S = single-pixel, M = multi-pixel block).

Value of k		BA-BLF	BA	BIOMASS	QMD	MEANDBH	STPH	DVPH	CANCOV	OGSI
$k = 1$	Kernel, Y, S	2.74	0.51	0.76	0.57	0.52	1.55	1.85	0.29	0.56
	Kernel, Y, M	2.43	0.44	0.66	0.49	0.45	1.37	1.61	0.25	0.49
	Kernel, N, S	2.68	0.45	0.68	0.50	0.48	1.45	1.76	0.26	0.52
	Kernel, N, M	2.43	0.41	0.62	0.45	0.43	1.31	1.59	0.23	0.47
	Single, Y, S	2.69	0.52	0.78	0.59	0.53	1.54	1.86	0.29	0.57
	Single, Y, M	2.28	0.43	0.64	0.47	0.43	1.32	1.54	0.24	0.47
	Single, N, S	2.61	0.47	0.70	0.52	0.49	1.49	1.80	0.26	0.53
	Single, N, M	2.28	0.39	0.59	0.43	0.41	1.27	1.51	0.21	0.45
$k = 5$	Kernel, Y, S	2.24	0.42	0.63	0.46	0.43	1.27	1.50	0.23	0.47
	Kernel, Y, M	2.14	0.40	0.60	0.43	0.40	1.23	1.44	0.23	0.44
	Kernel, N, S	2.20	0.38	0.59	0.42	0.41	1.23	1.47	0.21	0.44
	Kernel, N, M	2.16	0.37	0.56	0.40	0.39	1.19	1.42	0.20	0.42
	Single, Y, S	2.25	0.44	0.65	0.47	0.44	1.27	1.52	0.24	0.48
	Single, Y, M	2.10	0.39	0.58	0.43	0.40	1.21	1.41	0.22	0.43
	Single, N, S	2.20	0.39	0.60	0.43	0.41	1.22	1.48	0.21	0.45
	Single, N, M	2.12	0.36	0.55	0.40	0.38	1.17	1.39	0.20	0.42
Average of all maps		2.35	0.42	0.64	0.47	0.44	1.32	1.57	0.24	0.48

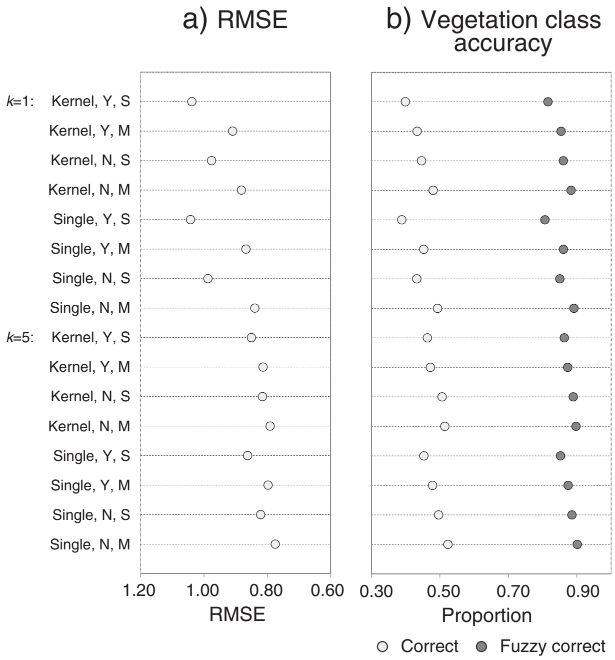

Fig. 4. Diagnostics from LOO cross-validation for GNN models developed with combinations of value of k, imputation grain (kernel or single-pixel), with (Y) and without (N) heterogeneous plots, and accuracy assessment grain (S = single-pixel, M = multi-pixel block). RMSEs are averages for nine continuous forest attributes (see Table 2). Vegetation class accuracy is overall accuracy (vegetation classes and 'fuzzy correct' classes are shown in Table 3).

relatively small influence on systematic agreement (AC_{SYS}) at the plot scale (Fig. 6).

3.4. Accuracy assessment grain

Values for map diagnostics differed strongly as a function of accuracy assessment grain, although it should be noted that these differences were associated with accuracy assessment methods rather than actual differences among maps. The RMSEs for continuous forest attributes and vegetation class accuracies were consistently better when assessed with a multi-pixel block than with a single pixel, for all scaling options (Fig. 4). At the plot scale, unsystematic agreement (AC_{UNS}) also was better when assessed with a multi-pixel block, but systematic agreement (AC_{SYS}) was slightly better when assessed with a single pixel (Figs. 6, 7a). For all diagnostics, accuracy differences between

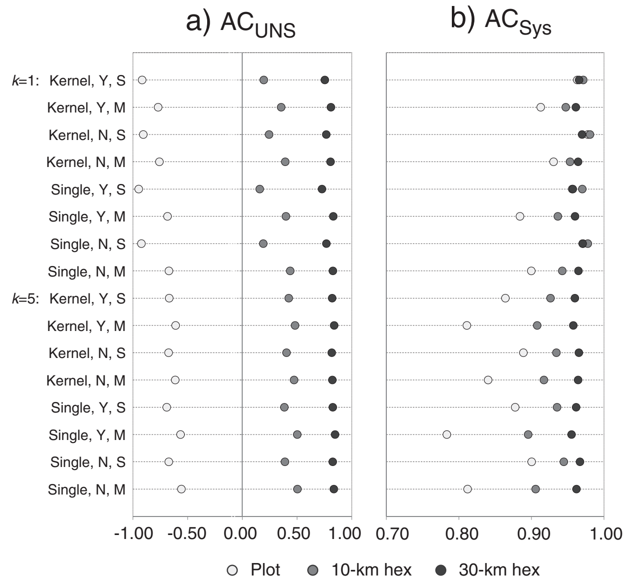

Fig. 6. Unsystematic (AC_{UNS}) and systematic (AC_{SYS}) agreement coefficients for GNN models developed with combinations of value of k, imputation grain (kernel or single-pixel), with (Y) and without (N) heterogeneous plots, and accuracy assessment grain (S = single-pixel, M = multi-pixel block). AC values are averages for nine continuous forest attributes (see Table 2), based on n = 1115 FIA Annual Inventory plots (including heterogeneous plots), and summarized for three spatial extents (plot, 10-km hexagon, and 30-km hexagon).

single- and multi-pixel accuracy assessment were greater for $k = 1$ than for $k = 5$, probably because $k = 5$ maps already reflect the averaging of multiple plot values.

3.5. Accuracy assessment extent (from plots to hexagons)

In general, differences in AC_{SYS} among the scaling options were small, and the differences diminished with increasing spatial extent (from plot to 10-km hexagon to 30-km hexagon) (Figs. 6 and 7). For $k = 1$ maps, systematic agreement (AC_{SYS}) was relatively consistent across spatial extents when assessed at the single-pixel scale, and improved very slightly with increasing extent when assessed with a multi-pixel block (Fig. 6). For $k = 5$ maps, AC_{SYS} was slightly better at larger spatial extents for both single- and multi-pixel accuracy assessment (Fig. 6). Fig. 7 illustrates, for the old-growth structure index (*OGSI*), how the geometric mean functional relationship (GMFR) approached the 1:1

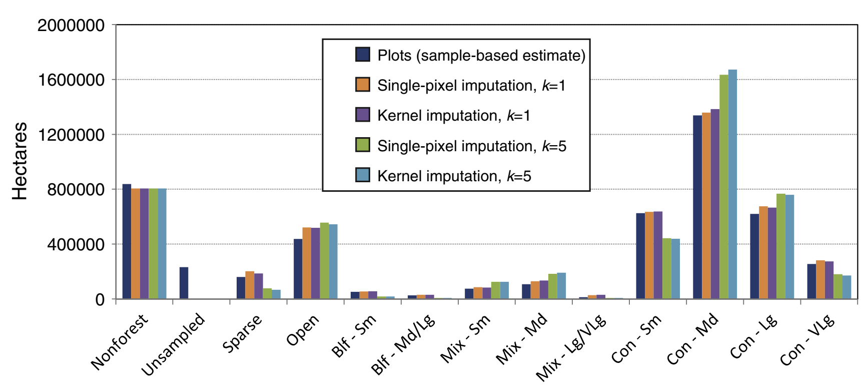

Fig. 5. Plot- and GNN-based estimates of forest area by vegetation class (*VEGCLASS*) (see Table 3 for definitions) for combinations of imputation grain and value of k. Heterogeneous plots were excluded. For GNN, the 'nonforest' area is derived from ancillary spatial data sources (nonforest mask). The 'unsampled' area, which represents plots not visited on the ground due to denied access or hazardous conditions, is shown here to provide a full accounting of the inventoried area.

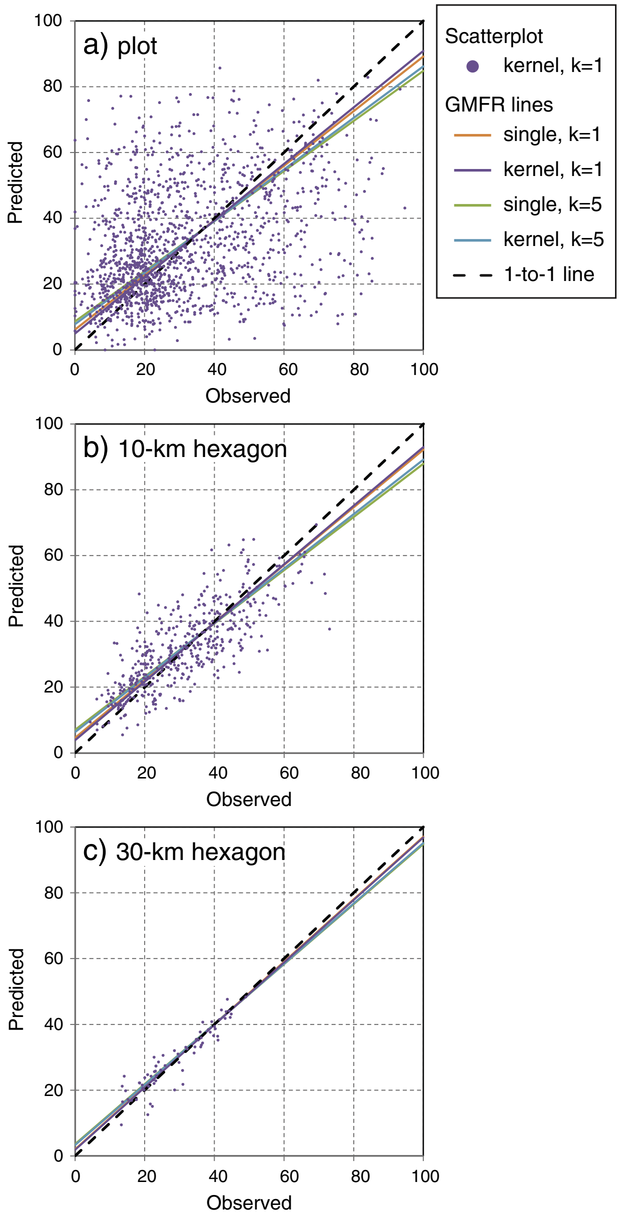

Fig. 7. Geometric mean functional relationships (GMFRs) between observed (plot) and predicted (GNN) estimates of old-growth structure index (*OGSI*) (see Table 2) for combinations of imputation grain and value of *k*, for three spatial extents: a) plot, b) 10-km hexagon, and c) 30-km hexagon. GNN models shown here excluded heterogeneous plots and predicted values were from a multi-pixel block. Scatter is shown for only one model for readability. Observed values are for n = 1115 FIA Annual Inventory plots.

line with increasing spatial extent for all maps assessed with a multi-pixel block.

Unsystematic agreement (AC_{UNS}) increased strongly with increasing spatial extent for all scaling combinations (Fig. 6). The trend of decreasing random error (less scatter around the GMFR) with spatial extent is shown graphically for OGSI and one scaling combination (kernel imputation, $k = 1$, multi-pixel accuracy assessment) (Fig. 7).

3.6. Value of k

For all scaling options, $k = 5$ maps were better than $k = 1$ maps for RMSE and AC_{UNS} for continuous forest attributes, and for overall vegetation class accuracy (Figs. 4 and 6). However, $k = 5$ maps were worse than $k = 1$ maps in terms of systematic agreement (AC_{SYS}) (Fig. 6) and for area distributions of vegetation classes (Fig. 5). Most notably, predicted area of vegetation classes from $k = 5$ deviated substantially from plot-based estimates for both the most and the least abundant vegetation classes. The $k = 5$ maps contained under-predictions of sparse, small broadleaf, and small conifer (early-successional) classes as well as very large conifer (late-successional), and over-prediction of the medium conifer class (mid-successional) (Fig. 5). The under-predicted classes were the least abundant in our study area and sampled by the fewest reference plots, whereas the over-predicted classes were most common and contained the most plots. As the exceptions to this pattern, the small and medium mixed classes were relatively rare in the landscape but also overpredicted with $k = 5$, which we attribute to the averaging of broadleaf proportion (*BA-BLF*) across multiple plots.

3.7. Map spatial pattern

The spatial pattern ("look and feel") of forest vegetation maps is important to many map users, and relates directly to forest fragmentation, habitat connectivity, and other measures of forest and landscape pattern. A quantitative evaluation of differences in spatial patterns among the scaling options is beyond the scope of this study. Nevertheless, we observed qualitatively that kernel and single-pixel imputation with $k = 1$ produced maps with overall patterns of forest conditions that were very similar to one another (Fig. 8). As one would expect, maps produced with single-pixel imputation contained more fine-scale variability within forest patches compared to kernel imputation. Any given forest patch appeared more homogenous (contained fewer pixels of contrasting forest condition) in the kernel imputation than in the single-pixel imputation. Maps constructed with $k = 5$ were even more smoothed in appearance, and the reduced areas of the more extreme vegetation classes (e.g., open or very large conifer) were readily apparent (Fig. 8). However, determining whether the spatial pattern and fine-scale variability in the spatial predictions are attributed to accurate depiction of actual ground conditions or just random prediction error is very difficult. Preferences of map users for particular spatial patterns are mostly subjective.

4. Discussion

4.1. Trade-offs between local prediction accuracy, range of variability, and areal representation

Many of the observed differences in map accuracy among the scaling options reflect various ways that averaging influences the imputation process and the calculation of diagnostics. Averaging takes place: (1) across pixels (values for spatial predictor variables) in kernel imputation, (2) across reference plots (response variables) in $k = 5$ imputation, (3) across pixels (predicted response variables) in multi-pixel accuracy assessment, and (4) across plots (both predicted and observed response variables) at larger spatial extents (10- and 30-km hexagons). Some of the differences in diagnostics among maps reflect real differences in the maps, while others are simply artifacts of the calculation process. To further complicate matters, the effects of averaging in different modeling and mapping steps can be compounded or cancel each other out. For example, for $k = 5$ the AC_{SYS} statistics were slightly different between the two imputation grains and the two accuracy assessment grains at the plot extent, but were nearly the same at the 30-km-hexagon extent.

In general though, averaging has a positive effect on local-scale prediction accuracy (RMSEs and vegetation class accuracy (Fig. 4), and AC_{UNS} (Fig. 6)). This is not surprising, as plots with quite similar spatial predictor values can have very different conditions on the ground (values for response variables). However, local accuracy gained by averaging comes at the expense of a loss of variability in the spatial

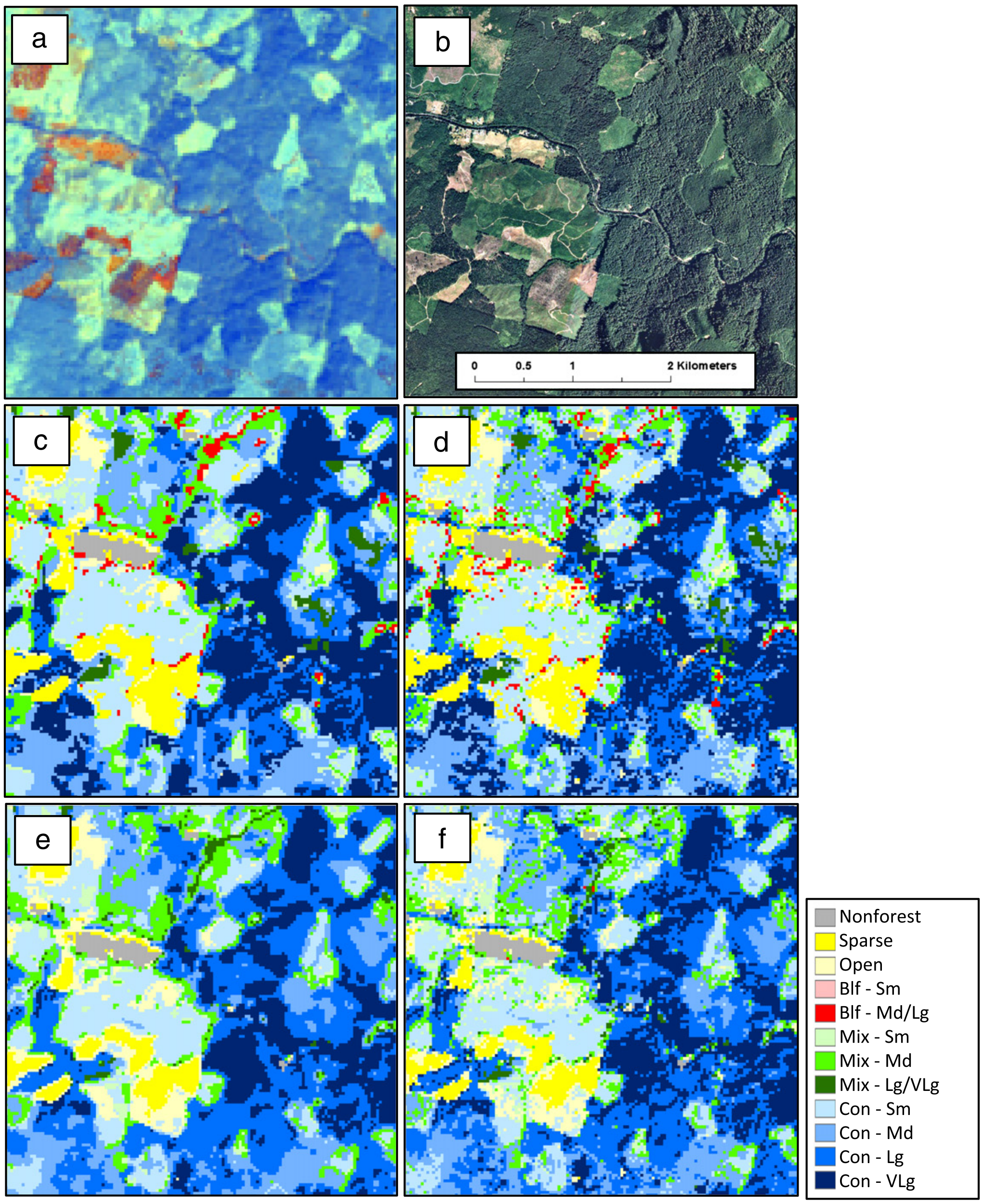

Fig. 8. Imagery and GNN maps for 2005. a) Landsat tasseled cap components brightness–greenness–wetness (shown as red–green–blue image stack). b) Digital aerial photo. c)–f) Vegetation class (*VEGCLASS*) maps (see Table 3) from GNN models with no heterogeneous plots: c) kernel imputation, $k = 1$; d) single-pixel imputation, $k = 1$; e) kernel imputation, $k = 5$; and f) single-pixel imputation, $k = 5$.

predictions across a broader region, as compared to the reference data. This trade-off has been well documented (e.g., LeMay & Temesgen, 2005; McRoberts, Nelson, & Wendt, 2002; Moeur & Stage, 1995; Ohmann & Gregory, 2002), and was an expected finding of this analysis.

Among the scaling options we evaluated, the averaging effect was most notable for value of k and for accuracy assessment grain. Vegetation maps based on $k = 1$ and $k = 5$ were quite different in appearance (Fig. 8) and in predicted area of forest conditions relative to plot-based estimates (Fig. 5). Systematic agreement (AC_{SYS}) was notably less for $k = 5$ maps at the plot scale (Fig. 6). Although k is not strictly speaking an aspect of spatial scale, imputing with $k > 1$ is effectively scaling up model predictions by averaging over multiple reference observations. The effect of averaging observations for multiple reference plots, whether across multiple pixels where $k = 1$ or for a single pixel when $k > 1$, was similar. We expect the effects of k would be even more pronounced for the much larger values of k that are commonly reported (Eskelson et al., 2009). Past studies have shown that small values of k result in reduced local prediction accuracy (greater RMSEs), but are more likely to produce estimates that may conserve more of the range of variability present in the plot sample, and are less likely to yield estimates that are beyond the bounds of reality (LeMay & Temesgen, 2005). In addition, our findings show that maps developed with $k = 1$ were notably better for rare forest conditions. The areas of early-successional (sparse), late-successional (very large conifer), and broadleaf vegetation classes in our

$k = 1$ maps very closely approximated the plot-based area estimates (Fig. 5). Less abundant forest types such as these often have disproportionately high values for wildlife habitat or other purposes, and their representation in regional maps can be quite important to map users.

Imputed maps based on $k > 1$ may result in unrealistic assemblages of species within map units, as demonstrated by Grossmann, Ohmann, Gregory, and May (2009). For example, species that do not normally co-occur in nature, such as a subalpine species and a low-elevation species, may be imputed to the same pixel. Ultimately, the optimal value of k will depend on the particular problem (objectives and data), and is a trade-off between the accuracy (RMSE) of the estimates and the variation that is retained in the estimates (McRoberts et al., 2002).

Differences in diagnostic values between single- and multi-pixel accuracy assessments reflected post-mapping averaging rather than real differences among the vegetation maps. The averaging of values for nine pixels yielded predicted values that were closer to the mean, and therefore more likely to have smaller deviations from observed values (lower RMSE and greater AC_{UNS}). However, predicted values from multi-pixel accuracy assessment had lower systematic agreement (AC_{SYS}) with observed values at the plot extent (Fig. 6). Again, the lower AC_{SYS} scores do not reflect real trends in the mapped predictions, but rather are an artifact of averaging in the calculation.

4.2. Implications for map developers and users

Our findings demonstrate that no single set of scaling methods yields optimal results across all measures of map accuracy. For both map developers and users, choices regarding spatial scale will hinge on study objectives and intended applications of the maps. Nevertheless, some general conclusions can be drawn. If the primary concern is a map with the best possible accuracy at the local (plot) scale, maps developed with $k > 1$ and single-pixel imputation consistently provided the best results for RMSE and AC_{UNS} (Figs. 4 and 6). But users of such maps need to keep in mind that regional-scale representation of the range of variability would be less reliable, and that the maps may contain pixels with unlikely combinations of entities, such as species that do not co-occur in nature (Grossmann et al., 2009). If a map application demands realistic representation of the range of variation of forest conditions over a broader region – including the occurrence of forest conditions at the 'tails' of the distributions (extreme values) of forest attributes – then maps developed with $k = 1$ may be a better choice. Users concerned with more reliable map portrayal of the co-occurrence of species and forest structures might also opt for a $k = 1$ map (Henderson, Ohmann, Gregory, Roberts, & Zald, in press). However, users also need to expect lower local-scale accuracy in such maps.

Inevitably, the features of an ideal spatial modeling framework must be tempered by limitations of data availability, computational demands, and funding. Given the availability of plot and geospatial data similar to ours, the scaling methods we compared are operationally quite similar with one exception: the human labor required to screen for heterogeneous plots. In deciding whether to undertake this endeavor, map developers need to weigh the benefits of outlier screening, in terms of potential improvement in map accuracy, against the costs in person-hours. We had hoped that heterogeneous plots could be retained in the kernel imputation maps without loss of accuracy, since a heterogeneous plot could be imputed to a kernel that spanned multiple conditions on the ground and in the imagery. However, this was not the case. Because our spatial predictor values for plot locations do not capture the spatial arrangement of forest conditions within a plot or kernel, a plot that is half young plantation and half old growth (for example) would have the same values for spatial predictors and response variables as a plot of sparse large trees. Thus we could not ensure that the plot(s) with the best match in forest conditions would be chosen as nearest neighbor(s).

Maps that included heterogeneous plots in their development exhibited more egregious errors in specific map locations where values for the heterogeneous plots were imputed, as revealed by visual inspection of outlier plots from the GNN maps in the imagery. The aggregate effect of these errors on accuracy at the regional level was small but noticeable, and could be important for some map applications. In situations where within-plot heterogeneity is recorded for all plots (such as for FIA Annual Inventory plots), map developers could choose to simply exclude from modeling all plots containing multiple forest conditions, hence avoiding the cost of outlier screening. If heterogeneous plots are to be excluded, we recommend careful scrutiny of the process to assure that rules are objectively and consistently applied, and to check that spatial predictions are not inadvertently biased.

By providing maps consisting of a large suite of forest attributes, at 30-m resolution, map developers run the risk of creating the illusion of precision in the maps. In fact, none of the alternative scaling methods we examined provided high map accuracy at the scale of the individual pixel or 3-by-3-pixel block, which is desired by many map users. These results are comparable to other Landsat-based mapping efforts in our region (e.g., Cohen, Maiersperger, Spies, & Oetter, 2001), and reflect the inherent limitations of the Landsat sensors and data for mapping forest conditions. Even maps developed from other imagery sources with finer spatial and spectral resolution will nevertheless differ from field measurements taken on the ground. Therefore, it is incumbent on map developers to provide diagnostics that characterize map reliability in a variety of ways, and to encourage map users to be aware of the limitations of the data.

For maps produced using any of the scaling methods we examined in this study – as well as other methods of predictive vegetation mapping that rely on Landsat data – users can minimize the risk of negative consequences of random error in the forest maps by summarizing the information to broader spatial extents. While both AC_{SYS} and AC_{UNS} increased with increasing extent from plot to 10- and 30-km hexagons, AC_{SYS} was generally quite good for all extents (Fig. 6).

In cases where the mapped forest attributes are not used directly, but rather are used to support subsequent modeling applications (e.g., for wildlife habitat suitability or forest health risk), we recommend the analyses be applied to the original 30-m data and the model results then summarized and interpreted at coarser grains or extents. This approach both preserves logical species (or structure) co-occurrence during analysis while providing results at a scale where the maps are more reliable.

A key advantage of nearest-neighbor imputation is the capability to simultaneously map multiple forest attributes – essentially anything measured on the reference (plot) data – in one step (in our case, with a single multivariate CCA model). Our intent with the GNN approach (CCA modeling followed by mapping using nearest-neighbor imputation) is to maintain the finest level of vegetation detail in the final maps that is practical (tree-level data at the grain of 30-m pixels), which provides many diverse users the greatest possible flexibility to summarize information to meet their particular objectives and analytical needs. While it may appear incongruent to provide multi-pixel accuracy assessment with single-pixel maps (Stehman & Wickham, 2011), we consistently counsel map users against relying on pixel-scale information in the maps. Effectively, our population of interest is 3-by-3 blocks of pixels (Figs. 2 and 3). Given map and plot data similar to ours, we therefore recommend that map developers calculate diagnostics for multi-pixel blocks, and document this fact in the map metadata for the benefit of map users. When evaluating maps for potential use, especially when comparing alternative maps, map users need to be aware of the potentially strong effect that accuracy assessment grain can have on results.

Although our analysis of scaling effects was conducted for one particular region, many of our findings should be widely applicable to other forested landscapes, datasets, and mapping methods. Our regional and national forest inventory plots are comparable to those in other regions and countries (McRoberts, 2009), and the Landsat archive is now accessible at no cost (Woodcock et al., 2008). Our study region encompasses long and strong gradients of climate and elevation, and a wide diversity of forest types (Ohmann & Spies, 1998). Other than the value of k, the scaling options we examined are not specific to nearest-neighbor imputation, and findings are generalizable to methods of predictive vegetation mapping based on the integration of plot and geospatial data. In addition, although our analysis was multi-temporal in terms of both plots and imagery, our approach is equally applicable to single-date analyses.

5. Conclusions

The integration of regional and national forest inventory plot data with remotely sensed data such as acquired by the Landsat sensors provides a powerful tool for mapping of forest vegetation conditions over large regions. However, several methods of geographic co-registration and combining values from the two data types for spatial modeling and accuracy assessment can have major effects on the resulting forest maps and diagnostics. We have presented a case study for a large region in the Pacific Northwest, USA that quantifies the effects of several aspects of spatial scale – mostly related to spatial resolution – on forest maps. Our findings show that no single set of scaling methods yields optimal results for all measures of map accuracy, although some general conclusions can be drawn. For both map developers and users, choices regarding spatial scale will hinge on study objectives and intended applications of the maps. By providing a variety of map diagnostics that are associated with a range of scaling options, both map developers and users can make informed choices about methods and resulting maps that best meet their particular objectives. Put in a different way, map users can use the diagnostics to evaluate the appropriateness of map products for their applications at different scales.

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Appendix A

User's accuracy, producer's accuracy, and kappa statistics for vegetation classes for GNN maps developed with different scaling options: value of k, imputation grain (kernel or single-pixel), with (Y = Yes, n = 8670) and without (N = No, n = 6036) heterogeneous plots, and accuracy assessment grain (S = single-pixel, M = multi-pixel block). See Table 3 for vegetation class definitions, abbreviations, and those considered to be 'fuzzy' correct.

Value of *k*		Sparse	Open	Blf-Sm	Blf-Md/Lg	Mix-Sm	Mix-Md	Mix-Lg/VLg	Con-Sm	Con-Md	Con-Lg	Con-VLg	All vegetation classes
Producer's accuracy (percentage correct)													
$k = 1$	Kernel, Y, S	34	50	33	12	7	10	0	33	48	38	29	40
	Kernel, Y, M	49	54	33	19	12	16	0	39	49	39	32	43
	Kernel, N, S	40	56	30	24	7	7	4	38	53	41	34	45
	Kernel, N, M	53	58	42	21	8	14	0	43	54	44	36	48
	Single, Y, S	35	48	25	10	10	10	0	33	47	38	27	39
	Single, Y, M	58	56	39	20	15	17	4	44	49	41	33	45
	Single, N, S	38	53	34	25	9	9	0	36	52	40	33	43
	Single, N, M	61	60	41	42	10	17	0	47	55	43	36	49
$k = 5$	Kernel, Y, S	72	57	33	21	11	19	0	44	50	43	36	46
	Kernel, Y, M	80	57	35	22	16	20	0	47	50	43	42	47
	Kernel, N, S	73	60	29	23	14	24	10	50	54	46	41	51
	Kernel, N, M	77	60	39	27	17	22	0	53	55	46	41	51
	Single, Y, S	62	55	34	29	11	19	0	42	50	42	36	45
	Single, Y, M	84	58	29	33	15	22	0	49	50	44	44	48
	Single, N, S	68	58	43	30	13	25	0	47	54	44	40	50
	Single, N, M	85	61	36	18	15	24	0	56	56	46	43	52
Producer's accuracy (percentage 'fuzzy' correct)													
$k = 1$	Kernel, Y, S	79	78	73	48	72	54	30	84	84	90	71	82
	Kernel, Y, M	90	83	73	61	77	61	28	89	85	93	79	85
	Kernel, N, S	85	84	75	62	77	59	44	89	86	93	79	86
	Kernel, N, M	92	85	75	47	81	65	42	92	88	96	80	88
	Single, Y, S	78	76	70	44	69	54	38	84	83	90	69	81
	Single, Y, M	94	84	82	48	83	65	50	90	85	94	78	86
	Single, N, S	85	82	83	57	75	55	38	87	85	93	79	85
	Single, N, M	94	87	84	50	79	63	46	93	88	97	82	89
$k = 5$	Kernel, Y, S	96	85	83	46	75	59	56	91	85	94	78	86
	Kernel, Y, M	98	86	75	52	83	62	43	93	85	95	84	87
	Kernel, N, S	97	87	89	46	81	66	60	93	88	97	81	89
	Kernel, N, M	98	88	94	45	89	70	50	94	88	96	83	90
	Single, Y, S	95	83	79	50	78	61	42	89	85	92	75	85
	Single, Y, M	98	87	71	52	84	64	43	92	85	95	84	88
	Single, N, S	98	86	100	40	73	65	57	93	87	96	84	89
	Single, N, M	100	89	86	36	83	70	100	94	88	97	82	90
User's accuracy (percentage correct)													
$k = 1$	Kernel, Y, S	29	48	30	12	7	9	0	34	49	37	30	40
	Kernel, Y, M	23	52	20	12	13	18	0	33	57	44	21	43
	Kernel, N, S	39	53	34	18	6	6	3	38	54	42	34	45
	Kernel, N, M	33	57	28	14	8	14	0	38	60	49	28	48
	Single, Y, S	34	45	24	10	10	9	0	34	47	38	29	39
	Single, Y, M	22	53	18	10	15	21	2	32	61	48	19	45
	Single, N, S	38	51	26	25	8	9	0	37	52	42	33	43
	Single, N, M	30	60	25	18	11	18	0	37	63	49	25	49

(continued on mext page)

Appendix A (*continued*)

Value of k		Sparse	Open	Blf-Sm	Blf-Md/Lg	Mix-Sm	Mix-Md	Mix-Lg/VLg	Con-Sm	Con-Md	Con-Lg	Con-VLg	All vegetation classes
User's accuracy (percentage correct)													
$k = 5$	Kernel, Y, S	18	55	14	10	11	25	0	31	62	50	22	46
	Kernel, Y, M	14	53	9	10	14	26	0	28	67	53	18	47
	Kernel, N, S	22	61	15	11	14	28	3	35	65	51	31	51
	Kernel, N, M	21	60	13	11	19	27	0	35	68	54	26	51
	Single, Y, S	18	54	14	14	12	23	0	30	60	49	25	45
	Single, Y, M	14	53	5	14	13	29	0	26	68	55	16	48
	Single, N, S	21	59	11	11	14	31	0	33	64	51	29	50
	Single, N, M	21	62	9	7	15	29	0	34	69	57	24	52
User's accuracy (percentage 'fuzzy' correct)													
$k = 1$	Kernel, Y, S	76	75	64	40	65	60	31	86	86	90	70	82
	Kernel, Y, M	77	73	68	48	71	71	47	90	90	94	79	85
	Kernel, N, S	83	80	77	57	70	61	31	89	88	94	81	86
	Kernel, N, M	87	79	81	57	67	69	41	92	90	96	86	88
	Single, Y, S	77	74	62	40	67	52	31	86	84	90	69	81
	Single, Y, M	78	71	68	46	68	75	49	91	92	96	79	86
	Single, N, S	84	79	72	39	71	61	31	87	88	93	79	85
	Single, N, M	88	77	81	54	80	77	59	93	92	97	87	89
$k = 5$	Kernel, Y, S	79	71	73	48	66	75	44	91	91	96	81	86
	Kernel, Y, M	80	67	72	50	61	79	58	92	94	97	85	87
	Kernel, N, S	88	76	81	64	71	80	48	92	91	98	88	89
	Kernel, N, M	88	75	83	68	62	81	62	93	93	98	90	90
	Single, Y, S	77	71	70	42	67	68	60	89	90	95	82	85
	Single, Y, M	80	66	72	52	59	79	60	93	94	97	84	88
	Single, N, S	87	75	81	50	70	72	48	92	92	97	87	89
	Single, N, M	88	74	83	61	67	80	62	94	93	98	91	90

Value of k		Sparse	Open	Blf-Sm	Blf-Md/Lg	Mix-Sm	Mix-Md	Mix-Lg/VLg	Con-Sm	Con-Md	Con-Lg	Con-VLg
Kappa statistics (correct)												
$k = 1$	Kernel, Y, S	0.29	0.40	0.31	0.11	0.06	0.07	−0.01	0.22	0.22	0.23	0.23
	Kernel, Y, M	0.30	0.45	0.25	0.14	0.11	0.14	0.00	0.26	0.26	0.27	0.20
	Kernel, N, S	0.37	0.46	0.31	0.20	0.05	0.05	0.03	0.29	0.29	0.27	0.27
	Kernel, N, M	0.39	0.50	0.33	0.17	0.07	0.12	0.00	0.32	0.33	0.32	0.26
	Single, Y, S	0.33	0.37	0.24	0.10	0.08	0.07	0.00	0.21	0.21	0.23	0.21
	Single, Y, M	0.30	0.47	0.24	0.13	0.14	0.17	0.02	0.28	0.28	0.30	0.19
	Single, N, S	0.36	0.43	0.29	0.25	0.08	0.07	0.00	0.26	0.27	0.26	0.26
	Single, N, M	0.38	0.53	0.30	0.25	0.09	0.16	0.00	0.34	0.35	0.32	0.24
$k = 5$	Kernel, Y, S	0.27	0.48	0.19	0.13	0.09	0.19	0.00	0.27	0.30	0.32	0.23
	Kernel, Y, M	0.22	0.47	0.15	0.13	0.13	0.20	0.00	0.27	0.30	0.34	0.21
	Kernel, N, S	0.33	0.53	0.19	0.14	0.13	0.24	0.05	0.34	0.36	0.35	0.30
	Kernel, N, M	0.32	0.53	0.19	0.15	0.17	0.23	0.00	0.35	0.38	0.36	0.26
	Single, Y, S	0.27	0.47	0.19	0.19	0.10	0.18	0.00	0.26	0.28	0.31	0.24
	Single, Y, M	0.23	0.48	0.09	0.19	0.12	0.23	0.00	0.26	0.31	0.35	0.20
	Single, N, S	0.31	0.51	0.18	0.16	0.12	0.26	0.00	0.31	0.35	0.33	0.28
	Single, N, M	0.33	0.55	0.15	0.10	0.14	0.24	0.00	0.35	0.39	0.38	0.26
Kappa statistics ('fuzzy' correct)												
$k = 1$	Kernel, Y, S	0.84	0.78	0.76	0.57	0.80	0.70	0.47	0.87	0.79	0.91	0.76
	Kernel, Y, M	0.87	0.78	0.79	0.66	0.84	0.76	0.55	0.91	0.82	0.94	0.85
	Kernel, N, S	0.88	0.82	0.83	0.70	0.84	0.73	0.53	0.91	0.82	0.94	0.84
	Kernel, N, M	0.92	0.82	0.85	0.65	0.84	0.78	0.58	0.94	0.84	0.96	0.88
	Single, Y, S	0.83	0.76	0.76	0.55	0.79	0.66	0.51	0.87	0.78	0.91	0.75
	Single, Y, M	0.88	0.77	0.81	0.60	0.84	0.79	0.65	0.92	0.82	0.95	0.85
	Single, N, S	0.89	0.81	0.84	0.58	0.83	0.71	0.50	0.89	0.82	0.94	0.84
	Single, N, M	0.93	0.82	0.88	0.63	0.88	0.80	0.71	0.94	0.85	0.97	0.89
$k = 5$	Kernel, Y, S	0.88	0.78	0.85	0.60	0.81	0.76	0.63	0.93	0.82	0.95	0.85
	Kernel, Y, M	0.89	0.76	0.82	0.64	0.81	0.78	0.71	0.94	0.83	0.96	0.89
	Kernel, N, S	0.93	0.81	0.90	0.71	0.85	0.81	0.67	0.94	0.84	0.97	0.88
	Kernel, N, M	0.94	0.81	0.91	0.74	0.85	0.83	0.76	0.95	0.85	0.97	0.91
	Single, Y, S	0.87	0.77	0.82	0.56	0.83	0.74	0.72	0.91	0.81	0.94	0.84
	Single, Y, M	0.88	0.76	0.82	0.64	0.81	0.79	0.73	0.94	0.83	0.96	0.89
	Single, N, S	0.93	0.81	0.91	0.60	0.82	0.76	0.67	0.94	0.84	0.96	0.89
	Single, N, M	0.93	0.81	0.90	0.68	0.84	0.82	0.78	0.95	0.86	0.98	0.92

Appendix B. Supplementary data

Supplementary data associated with this article can be found in the online version, at http://dx.doi.org/10.1016/j.rse.2013.08.048. These data include Google maps of the most important areas described in this article.

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